

15.4) 식 (15.11)을 증명하라.

$$E[\tilde{\sigma}^2] = \sigma^2 + \beta_r' X_r' [I - X_p(X_p' X_p)^{-1} X_p'] X_r \beta_r / (n-p-1)$$

$$X = [X_p, X_r]$$

$$y = X_p \beta_p + X_r \beta_r + \epsilon$$

X_p : 설명변과 p 개의 독립변수를 포함

X_r : $k-p$ 개의 독립변수를 포함

$$\tilde{\sigma}^2 = y' [I - X_p(X_p' X_p)^{-1} X_p'] y / (n-p-1)$$

$$\text{Reduced Model: } y = X_p \beta_p + \epsilon$$

$$\text{Full Model: } y = X_p \beta_p + X_r \beta_r + \epsilon$$

$$\tilde{\sigma}^2 = \text{MSE}(R)$$

$$E[\tilde{\sigma}^2] = \frac{1}{n-p-1} E[y' A y], \quad A = I - X_p(X_p' X_p)^{-1} X_p'$$

$$E[y' A y] = \text{trace}(A \Sigma) + \mu' A \mu, \quad \Sigma = \text{Var}(y)$$

$$= \sigma^2 \text{trace}[I - X_p(X_p' X_p)^{-1} X_p'] + (X \beta)' [I - X_p(X_p' X_p)^{-1} X_p'] (X \beta)$$

$$= \sigma^2(n-p-1) + (X_p \beta_p + \Pi_{X_p} X_r \beta_r + \Pi_{X_r^\perp} X_r \beta_r)' [I - X_p(X_p' X_p)^{-1} X_p'] (X_p \beta_p + \Pi_{X_p} X_r \beta_r + \Pi_{X_r^\perp} X_r \beta_r)$$

$$= \sigma^2(n-p-1) + (\Pi_{X_r^\perp} X_r \beta_r)' (I - \Pi_{X_p}) (\Pi_{X_r^\perp} X_r \beta_r)$$

$$= \sigma^2(n-p-1) + \{ (X_r - \Pi_{X_p} X_r) \beta_r \}' (I - \Pi_{X_p}) \{ (X_r - \Pi_{X_p} X_r) \beta_r \}$$

$$= \sigma^2(n-p-1) + \beta_r' X_r' (I - \Pi_{X_p}) X_r \beta_r$$

$$\therefore E[\tilde{\sigma}^2] = \sigma^2 + \frac{\beta_r' X_r' (I - \Pi_{X_p}) X_r \beta_r}{(n-p-1)} \quad \square$$

Q15.5) 정리 (15.1)을 증명하라.

《정리 15.1》 행렬 $\text{Var}(b_p) - \text{Var}(\tilde{\beta}_p)$ 은 양반정지행렬 (positive semi-definite matrix) 이다.

$$\begin{aligned} \hat{y} &= X_p b_p + X_r b_r = \Pi_{X_p} y + \Pi_{X_r^\perp} y \\ &= X_p b_p + \Pi_{X_p} X_r b_r + \Pi_{X_r^\perp} X_r b_r \end{aligned}$$

$$\therefore \Pi_{X_p} y - \Pi_{X_p} X_r b_r = X_p b_p$$

$$\begin{aligned} b_p &= (X_p' X_p)^{-1} X_p' y + (X_p' X_p)^{-1} X_p' X_r b_r \\ &= \tilde{\beta}_p + A b_r, \quad A = (X_p' X_p)^{-1} X_p' X_r \\ &\quad \tilde{\beta}_p = (X_p' X_p)^{-1} X_p' y \end{aligned}$$

$$\begin{aligned} \text{Var}(b_p) &= \text{Var}(\tilde{\beta}_p + A b_r) \\ &= \text{Var}(\tilde{\beta}_p) + A \text{Var}(b_r) A' + 2 \text{Cov}(\tilde{\beta}_p, A b_r) \end{aligned}$$

$$\text{Cov}(\tilde{\beta}_p, A b_r) = 0 \quad \because \tilde{\beta}_p \in C(X_p), \quad b_r \in C(X_p)^\perp$$

$$\therefore \text{Var}(b_p) = \text{Var}(\tilde{\beta}_p) + A \text{Var}(b_r) A'$$

$$\text{Var}(b_p) - \text{Var}(\tilde{\beta}_p) = A \text{Var}(b_r) A' = A \Pi \Pi' A'$$

$$\begin{aligned} \text{let } x \in \mathbb{R}^{p+1}, \quad x' \{ \text{Var}(b_p) - \text{Var}(\tilde{\beta}_p) \} x &= x' A \Pi \Pi' A' x \\ &= \| \Lambda^{1/2} \Pi' A' x \|^2 \geq 0 \end{aligned}$$

$\therefore \text{Var}(b_p) - \text{Var}(\tilde{\beta}_p)$ is positive semi-definite.

15.6) 정리 (15.2)를 증명하라.

$$\langle\langle \text{정리 15.2} \rangle\rangle \quad \text{Var}(\hat{y}) \geq \text{Var}(\tilde{y}_p)$$

$$\hat{y} = X(X'X)^{-1}X'y$$

$$\tilde{y}_p = X_p(X_p'X_p)^{-1}X_p'y$$

$$\text{Var}(\hat{y}) = \Pi_X \text{Var}(y) \Pi_X' = \sigma^2 \Pi_X = \sigma^2 X(X'X)^{-1}X'$$

$$\text{Var}(\tilde{y}_p) = \Pi_{X_p} \text{Var}(y) \Pi_{X_p}' = \sigma^2 \Pi_{X_p}$$

$$\text{Var}(\hat{y}) - \text{Var}(\tilde{y}_p) = \sigma^2(\Pi_X - \Pi_{X_p}) = \sigma^2 \Pi_{X_{r,\perp}}$$

$$\begin{aligned} \Pi_{X_{r,\perp}} &= X_{r,\perp}(X_{r,\perp}'X_{r,\perp})^{-1}X_{r,\perp}' \\ X_{r,\perp} &= X_r - \Pi_{X_p}X_r \end{aligned}$$

Suppose $c \in \mathbb{R}^n$ is an arbitrary vector

$$\begin{aligned} c'(\sigma^2 \Pi_{X_{r,\perp}})c &= \sigma^2 [c' \Pi_{X_{r,\perp}}] (\Pi_{X_{r,\perp}} c) \\ &= \|\sigma \Pi_{X_{r,\perp}} c\|^2 \geq 0 \end{aligned}$$

$\therefore \text{Var}(\hat{y}) - \text{Var}(\tilde{y}_p)$ is a positive semi-definite matrix.

15.7) 정리 (15.3)를 증명하라.

$\langle\langle \text{정리 15.3} \rangle\rangle$ 만약 행렬 $\text{Var}(b_r) - \beta_r \beta_r'$ 이 양반정지행렬이면

$\text{Var}(b_p) - \text{MSE}(\tilde{y}_p)$ 도 양반 정지행렬이며,

$\text{Var}(\hat{y}) \geq \text{MSE}(\tilde{y}_p)$ 이 성립한다.

$$\begin{aligned} \hat{y} &= X_p b_p + X_r b_r = \Pi_{X_p} y + \Pi_{X_{r,\perp}} y \\ &\quad \downarrow \\ &= X_p (X_p' X_p)^{-1} X_p' y + X_{r,\perp} (X_{r,\perp}' X_{r,\perp})^{-1} X_{r,\perp}' y \\ &\quad \downarrow \\ &= X_p b_p + \Pi_{X_p} X_r b_r + \Pi_{X_{r,\perp}} X_r b_r \end{aligned}$$

$$X_p b_p + \Pi_{X_p} X_r b_r = \Pi_{X_p} y$$

$$\therefore b_p = (X_p' X_p)^{-1} X_p' y - (X_p' X_p)^{-1} X_p' X_r b_r$$

$$\text{Var}(b_p) = \text{Var}((X_p' X_p)^{-1} X_p' y - (X_p' X_p)^{-1} X_p' X_r b_r) \quad , \quad b_r = (X_{r,\perp}' X_{r,\perp})^{-1} X_{r,\perp}' y$$

$$\begin{aligned} \text{MSE}(\tilde{y}_p) &= E[(\tilde{y}_p - \beta_p)^2] = E[(\tilde{y}_p - E[\tilde{y}_p] + E[\tilde{y}_p] - \beta_p)^2] \\ &= E[(\tilde{y}_p - E[\tilde{y}_p])^2] + E[(\text{bias}(\tilde{y}_p))^2] + 2E[(\tilde{y}_p - E[\tilde{y}_p])(E[\tilde{y}_p] - \beta_p)'] \\ &= \text{Var}(\tilde{y}_p) + \text{bias}(\tilde{y}_p) \text{bias}(\tilde{y}_p)' \end{aligned}$$

$$\text{bias}(\tilde{y}_p) = E[\tilde{y}_p] - \beta_p = E[b_p + A b_r] - \beta_p = A b_r \quad , \quad A = (X_p' X_p)^{-1} X_p' X_r$$

$$\therefore \text{bias}(\tilde{y}_p) \text{bias}(\tilde{y}_p)' = A b_r b_r' A'$$

$$\begin{aligned} \text{Var}(b_p) &= \text{Var}(\tilde{y}_p - A b_r) = \text{Var}(\tilde{y}_p) + A \text{Var}(b_r) A' - 2 \text{Cov}(\tilde{y}_p, b_r) \\ &= \text{Var}(\tilde{y}_p) + A \text{Var}(b_r) A' \quad (\because \tilde{y}_p \in C(X_p), b_r \in C(X_p)^\perp) \end{aligned}$$

$$\begin{aligned} \text{Var}(b_p) - \text{MSE}(\tilde{y}_p) &= \text{Var}(\tilde{y}_p) + A \text{Var}(b_r) A' - \text{Var}(\tilde{y}_p) - A \beta_r \beta_r' A' \\ &= A (\text{Var}(b_r) - \beta_r \beta_r') A' \geq 0 \end{aligned}$$

$\therefore \text{Var}(b_r) - \beta_r \beta_r'$ is semi-positive definite.

15.8) 식 (15.36)를 증명하라.

$$B = \int_R [\text{Var}(\hat{y}_j) - \text{MSE}(\tilde{y}_p)] dW(r) = \frac{1}{n} \sum_{j=1}^n [\text{Var}(\hat{y}_j) - \text{MSE}(\tilde{y}_p)] \quad \dots (15.35)$$

만약 $\text{Var}(b_r) = B \sigma^2$ 라 놓으면, 식 (15.35)는 다음과 같이 된다.

$$\frac{1}{n} \sum_{j=1}^n [\text{Var}(\hat{y}_j) - \text{MSE}(\tilde{y}_p)] = \frac{r}{n} \sigma^2 (I - \frac{\beta_r' \beta_r}{r \sigma^2}) \quad \dots (15.36)$$

* y_j : 실재의 측정값

* $\tilde{y}_{pj}$: 회귀방정식에 의한 추정값. $\tilde{y}_p = X_p \tilde{\beta}_p$

$$\frac{1}{n} \sum_{j=1}^n [\text{Var}(\hat{y}_j) - \text{MSE}(\tilde{y}_p)]$$

$$\begin{aligned} &= \frac{1}{n} \{ \text{trace}(\text{Var}(y)) - \text{trace}(\text{MSE}(\hat{y}_p)) \} \\ &= \frac{1}{n} \{ \text{trace}(\sigma^2 I_n) - \text{trace}(\text{Var}(\hat{y}_p) + \text{Bias}(\hat{y}_p) \text{Bias}(\hat{y}_p)') \} \end{aligned}$$

$$* \hat{y}_p = X_p (X_p' X_p)^{-1} X_p' y$$

$$\text{Var}(\hat{y}_p) = \sigma^2 (X_p' X_p)^{-1}$$

$$\begin{aligned} \text{trace}(\hat{y}_p) &= \sigma^2 \text{trace}(X_p' X_p)^{-1} \\ &= \sigma^2 \text{rank}(X_p' X_p) = \sigma^2 p \end{aligned}$$

$$\begin{aligned} \text{Bias}(\hat{y}_p) &= E[y] - E[\hat{y}_p] = X_p \beta_r + X_r \beta_r - E[X_p y] \\ &= X_p \beta_r + X_r \beta_r - \Pi_{X_p} (X_p \beta_p + X_r \beta_r) \\ &= (I_n - \Pi_{X_p}) X_r \beta_r \end{aligned}$$

$$\begin{aligned} \text{Bias}(\hat{y}_p) \text{Bias}(\hat{y}_p)' &= [(I_n - \Pi_{X_p}) X_r \beta_r] [(I_n - \Pi_{X_p}) X_r \beta_r]' \\ \text{trace}(\text{Bias}(\hat{y}_p) \text{Bias}(\hat{y}_p)') &= \text{trace}(\beta_r' X_r' (I_n - \Pi_{X_p}) X_r \beta_r) \\ &= \beta_r' X_r' (I_n - \Pi_{X_p}) X_r \beta_r \\ &= \beta_r' X_r' \Pi_{X_{r,\perp}} X_r \beta_r \\ &= \beta_r' X_{r,\perp}' \Pi_{X_{r,\perp}} X_{r,\perp} \beta_r \\ &= \beta_r' X_{r,\perp}' X_{r,\perp} (X_{r,\perp}' X_{r,\perp})^{-1} X_{r,\perp}' X_{r,\perp} \beta_r \\ &= \beta_r' (X_{r,\perp}' X_{r,\perp}) \beta_r \end{aligned}$$

$$\begin{aligned} \text{Var}(b_r) &= \text{Var}((X_{r,\perp}' X_{r,\perp})^{-1} X_{r,\perp}' y) \\ &= \sigma^2 (X_{r,\perp}' X_{r,\perp})^{-1} \end{aligned}$$

$$\begin{aligned} B \sigma^2 &= \text{Var}(b_r) \quad \therefore B = (X_{r,\perp}' X_{r,\perp})^{-1} \quad \therefore B^{-1} = X_{r,\perp}' X_{r,\perp} \\ &\quad \therefore \beta_r' B^{-1} \beta_r = \beta_r' (X_{r,\perp}' X_{r,\perp}) \beta_r \\ &= \text{trace}(\text{Bias}(\hat{y}_p) \text{Bias}(\hat{y}_p)') \end{aligned}$$

$$\frac{1}{n} \{ \text{trace}(\sigma^2 I_n) - \text{trace}(\text{Var}(\hat{y}_p) + \text{Bias}(\hat{y}_p) \text{Bias}(\hat{y}_p)') \}$$

$$\begin{aligned} &= \frac{1}{n} \{ n \sigma^2 - p \sigma^2 - \beta_r' B^{-1} \beta_r \} \\ &= \frac{\sigma^2 r}{n} \left\{ 1 - \frac{\beta_r' B^{-1} \beta_r}{r \sigma^2} \right\} \quad \checkmark \end{aligned}$$

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15.9.) 다음을 유도하라.

$$Q = \int_R [\text{Var}(\hat{y}) - \text{MSE}(\hat{y}_p)] dW(r)$$

$$= \sigma^2 \{ \text{tr}[(X'X)^{-1}M] - \text{tr}[(X_p'X_p)^{-1}M_{pp}] \} - \beta_r' [A'M_{pp}A - 2A'M_{pr} + M_{rr}] \beta_r$$

$$* M = \int_R x x' dW(r), \quad M_{ij} = \int_R x_i x_j' dW(r)$$

$$Q = \int_R [\text{Var}(\hat{y}) - \text{MSE}(\hat{y}_p)] dW(r)$$

$$= \int_R [\sigma^2 x'(X'X)^{-1}x - x_p'(X_p'X_p)^{-1}x_p \cdot \sigma^2 - \beta_r' [(x_p'A - x_r')(x_p'A - x_r')] \beta]$$

p.505

★ 15.10) 뒤로부터 제거하는 방법 (backward elimination method)에서

부분 F-검정치

$$F_o = \frac{b_1^2}{\text{Var}(b_1)}$$

이 15.2.2에 설명되어 있는 부분 F검정치

$$F_o = \frac{\text{SSE}(R) - \text{SSE}(F)}{\text{MSE}}$$

와 같음을 증명하라.

$$\text{SSE}(R) - \text{SSE}(F) = (A\hat{\beta})' (A(X'X)^{-1}A') (A\hat{\beta})$$

$$A = e_i' \text{ Let } A(X'X)^{-1}A' = e_i'(X'X)^{-1}e_i = c_{ii}$$

$$\text{Var}(b_i) = \text{Var}(e_i'b) = e_i' \text{Var}(b) e_i = \sigma^2 e_i'(X'X)^{-1} e_i$$

$$= \sigma^2 c_{ii}$$

$$\text{MSE} = \text{SST}/(n-p-1) = y'(I_n - \Pi_X)y / (n-p-1)$$

$$= \hat{\sigma}^2$$

$$\therefore \widehat{\text{Var}(b_i)} = \text{MSE} \cdot c_{ii} \Leftrightarrow \text{MSE} = \widehat{\text{Var}(b_i)} \cdot c_{ii}^{-1}$$

$$\therefore \frac{\text{SSE}(R) - \text{SSE}(F)}{\text{MSE}} = \frac{(A\hat{\beta})' (A(X'X)^{-1}A') (A\hat{\beta})}{\widehat{\text{Var}(b_i)} \cdot c_{ii}^{-1}}$$

$$= \frac{b_i \cdot c_{ii}^{-1} \cdot b_i}{\widehat{\text{Var}(b_i)} \cdot c_{ii}^{-1}} = \frac{b_i^2}{\widehat{\text{Var}(b_i)}}$$